Ali Ansari

Jaipur, Rajasthan

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Education

The LNM Institute of Information Technology, Jaipur *Expected May 2027*

B.Tech in Electronics & Communication Engineering

Ranked 1st out of 142 students for 6 semesters. | CGPA: 8.70

Projects

PitchLens | *Python, LangGraph, FAISS, BM25, Gemini Vision, FastAPI, Docker* | GitHub

- Built a **multimodal RAG** system over startup pitch decks; found 65% of pages across 19 real decks carried no PDF text layer, and built a per-page adaptive extractor routing image slides to **Gemini Vision**, lifting deck coverage from 4/19 to 19/19 (**4.75**×).
- Built **hybrid retrieval** (**BM25** + **FAISS** fused via reciprocal rank fusion, cross-encoder re-ranked), improving Recall@3 from 0.78 to 0.95 and MRR from 0.74 to 0.92 on a 72-question hand-labelled benchmark.
- Designed retrieval behind a single **Retriever** interface with a re-ranking decorator and a corpus-fingerprint cache cutting cold start from 3.9s to 0.036s (**109**×); served via **FastAPI** with a **LangGraph** corrective retrieve-grade-rewrite loop.

Micromouse Bot | *C++, Raspberry Pi Pico, PID Control, Flood Fill* | GitHub

- Built a tiny 10cm by 8cm maze-solving robot using the **Flood Fill algorithm** on **Raspberry Pi Pico W**.
- Used 3 **Infrared Sensors** for wall detection, **MPU6050** module for gyroscope data, and N20 Motors with Encoders for bot movement.
- Reduced IR sensor reading time from **300µs (official library) to 20µs** by writing custom **embedded C++** sensor code.
- Used **PID Control Algorithm** for precise $90 \pm 0.1°$ turns and smooth navigation.
- Used **interrupts**, **system timers**, and **multicore programming** for efficient sensor and control loops; secured 1st place in 3 National Micromouse competitions, including IIT Roorkee Cognizance '25, DTU Invictus '25, and BITS Pilani Apogee '25.

VoidCart | *MongoDB, Express.js, React, Node.js, Redux Toolkit, JWT, Cloudinary, Razorpay* | GitHub

- Built a full-stack **e-commerce platform** with a unified **admin dashboard**, user profiles, and order history tracking.
- Implemented **JWT-based authentication** and **Redux Toolkit** for cart state management on the **React** frontend.
- Integrated **Cloudinary** for product image uploads via **Multer** and **Razorpay** for payments; configured **Express.js** to serve the production React build for single-service deployment on **Render**.

Experience

Teaching Assistant – Programming for Problem Solving Lab Aug 2024 – Nov 2024

The LNM Institute of Information Technology, Jaipur

- Guided and assisted first-year students with **programming concepts** and lab assignments.
- Contributed 3+ hours weekly to help students **debug** and **optimize code** during lab sessions.

Achievements

- Rated **1404** (Specialist) on Codeforces, **1654** on CodeChef, **1863** (Knight) on Leetcode
- Global Rank **1293** on Codeforces Round 1017.
- Solved 1000+ problems across various coding platforms.

Technical Skills

Languages: C++, Python, SQL, JavaScript
Web Development: React, Node.js, Express.js, HTML, CSS, TailwindCSS
Databases: MongoDB, SQL
Tools: Git, GitHub, Postman, Cloudinary, Docker
CS Fundamentals: Data Structures and Algorithms, Object-Oriented Design, Database Management, Computer Networks, Operating Systems, Artificial Intelligence and Machine Learning.

Positions of Responsibility

Phoenix Club – Robotics Club of LNMIIT Dec 2023 - Aug 2025

Member

- Co-managed **Roborace** and **Robosoccer** during the technical fest, attracting 1000+ participants; designed event challenges and problem statements, collaborating with the core team to run them on **Unstop**.